LESSON PLAN

ROCKET ENGINE (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Types of fibers
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Types of fibers. Properties and uses of Glass fibers, Carbon fibers, Ceramics fibers, Boron fibers and Aramid fibers.
5. Specific Objectives	17.1.1.1 Trainee will be able to understand the development of composite materials, types and functions of fibers and resin, Defects in composite, selection criteria for Matrix, fabric weaves, prepress, strength flowcharts

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of fiber property charts; Instructor supervision required for safety regarding fiber dust inhalation and eye protection.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the fiber in a composite material?
(b) Name one common matrix material used in composites.
(c) Why are fibers essential for structural strength in aerospace applications?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to composite material development • Function of fibers and resin in composites • Overview of defects in composite structures • Selection criteria for matrix materials • Fabric weaves and prepress techniques	L & Ds	PPT, IB	15 min
(b)	• Properties and uses of Glass fibers • Properties and uses of Carbon fibers • Properties and uses of Ceramics fibers • Properties and uses of Boron fibers • Properties and uses of Aramid fibers	L & Ds	PPT, PDF	20 min
(c)	• Comparison of strength-to-weight ratios • Application of strength flowcharts • Summary of fiber selection criteria • Review of fabric weave patterns • Prepress requirements for different fibers	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Which fiber type offers the highest specific strength for aerospace structures?
(b) Explain the primary difference between Glass and Aramid fibers regarding impact resistance.
(c) What is the main selection criterion for choosing a matrix material for high-temperature applications?

12. Home Assignment and Follow Up	Prepare a brief note comparing the mechanical properties of Carbon fibers and Boron fibers to be discussed in the next lesson on matrix selection criteria.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Selection criteria for matrix material. Different types of matrix material and brief description of Thermosetting matrix material and Thermoplastic matrix material.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min